System Setup

We will dive right into coding in the very first lecture. You have different option on how to prepare your development environment for the class.

A: Install Git and Clone the Class Repository

All class material (lecture notes, exercise notebooks, dataset, etc.) will be distributed via a git repository.

Git clients are available for all relevant OS. You can use the official installers (https://git-scm.com/). If you are running Windows, TortoiseGit is also a good option (https://tortoisegit.org/).

Open a terminal and change into the directory you want to store the class files in (the <class directory>).

Depending on your GitLab setup (1FA, 2FA, ssh, PAT), use either

```
git clone git@inf-git.th-rosenheim.de:data_master/data_master_wise26.git
```

or

```
git clone https://inf-git.th-rosenheim.de/data_master/data_master_wise26.git
```

to clone the class repository.

If you haven't set up access to our GitLab `inf-git.th-rosenheim.de`, use the documentation at https://docs.gitlab.com. I would recommend setting up ssh-based access.

From now on, you can simply use

```
git pull
```

to get the latest class materials.

B: Setup Jupyter Environment

We will be working extensively with Jupyter Notebooks, so you need an environment to execute them in. All notebooks have been tested with defined versions of Python and the relevant libraries, so you should make sure to use a system with these exact same versions. The recommendation for this class is to use a docker-based setup, but if you want to, you can use local install or a cloud-based setup (version control depends on the cloud provider) as well.

Here are the steps for the docker-based environment (you only need to do these one time).

a. **Install Docker:** See https://docs.docker.com/install/
   For windows, using WSL2 is highly recommended (install WSL2 first, and setup docker to use it).

b. **Test the docker installation**
   Test your docker installation by running the Docker Hello World container with `docker run hello-world`:

   ```
   docker run hello-world
   ```

   Note: You may have to start the demon "Docker Desktop" or "Docker Demon" first.

c. **Build the docker image**

   For this class, we will be using a docker container based on the official `jupyter/scipy-notebook` hosted on `quay.io`. Unfortunately, it does not include all libraries we will be using (e.g., XGBoost), thus you will find a `Dockerfile` in the git repository, which uses it as a base and installs the missing libraries.

   Run

   ```
   docker build -t data .
   ```

   from the directory where you put the `Dockerfile` to build the `data` image (do not forget the trailing "dot" in the command!)

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C: Run the docker container

You need to start the docker container every time you want to work on the notebooks.

- As you cannot store documents inside a docker container (containers/images get destroyed all the time), you will have to "map" a persistent directory to store your files in into the container. We will be using the <class directory> as persistent directory and map it into the container as "/home/jovyan/work" (this path is predefined by the container) with the `-v` option.
- We also need to specify a port for the web-server to listen to, we will be using 8888 (`-p`),
- it makes sense to clean up the filesystem after running the container (`--rm`),
- and to give it a clear name (`--name`).

Examples:

Open a terminal, change into your <class directory> and run the container using

- On Linux or MacOS, execute

```
docker run -p 8888:8888 -v "$PWD":/home/jovyan/work --rm --name data data
```

- On Windows (e.g. storing your files in `C:\temp\`. replace this path if you are storing them in a different directory), execute

```
docker run -p 8888:8888 -v C:\temp:/home/jovyan/work --rm --name data data
```

If this works properly, it will show you how to access the Notebook-Server through a web browser, this should look like this (your token will be different every time you start the Notebook-Server):

```
To access the server, open this file in a browser:
      file:///home/jovyan/.local/share/jupyter/runtime/jpserver-7-open.html
  Or copy and paste one of these URLs:
      http://ee4d6332eca1:8888/?token=...
   or http://127.0.0.1:8888/?token=...
```

Copy-and-Paste the last URL (the one starting with `http://127.0.0.1:8888/`) into a web browser on your computer. You should now see the JupyterLab UI.

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